DIVIDED POWER STRUCTURE ON THE FIRST-ORDER THICKENING A \\OPLUS M

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file crystalline.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 07GL: sources/stacks-chow/tags.txt, source lines 9521–9521. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let (A, I, γ) be a divided power ring. Let M be an A-module. Let $B = A \oplus M$ as an A-algebra where M is an ideal of square zero and set $J = I \oplus M$. Set*

$$\delta_n(x + z) = \gamma_n(x) + \gamma_{n-1}(x)z$$

for $x \in I$ and $z \in M$. Then δ is a divided power structure and $A \to B$ is a homomorphism of divided power rings from (A, I, γ) to (B, J, δ).

Proof. We have to check conditions (1) – (5) of Divided Power Algebra, Definition 07GL. We will prove this directly for this case, but please see the proof of the next lemma for a method which avoids calculations. Conditions (1) and (3) are clear. Condition (2) follows from

$$\begin{aligned}
\delta_n(x + z)\delta_m(x + z) &= (\gamma_n(x) + \gamma_{n-1}(x)z)(\gamma_m(x) + \gamma_{m-1}(x)z) \\
&= \gamma_n(x)\gamma_m(x) + \gamma_n(x)\gamma_{m-1}(x)z + \gamma_{n-1}(x)\gamma_m(x)z \\
&= \frac{(n+m)!}{n!m!}\gamma_{n+m}(x) + \left(\frac{(n+m-1)!}{n!(m-1)!} + \frac{(n+m-1)!}{(n-1)!m!}\right)\gamma_{n+m-1}(x)z \\
&= \frac{(n+m)!}{n!m!}\delta_{n+m}(x + z)
\end{aligned}$$

Condition (5) follows from

$$\begin{aligned}
\delta_n(\delta_m(x + z)) &= \delta_n(\gamma_m(x) + \gamma_{m-1}(x)z) \\
&= \gamma_n(\gamma_m(x)) + \gamma_{n-1}(\gamma_m(x))\gamma_{m-1}(x)z \\
&= \frac{(nm)!}{n!(m!)^n}\gamma_{nm}(x) + \frac{((n-1)m)!}{(n-1)!(m!)^{n-1}}\gamma_{(n-1)m}(x)\gamma_{m-1}(x)z \\
&= \frac{(nm)!}{n!(m!)^n}(\gamma_{nm}(x) + \gamma_{nm-1}(x)z)
\end{aligned}$$

by elementary number theory. To prove (4) we have to see that

$$\delta_n(x + x' + z + z') = \gamma_n(x + x') + \gamma_{n-1}(x + x')(z + z')$$

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is equal to

$$\sum\nolimits_{i=0}^{n} (\gamma_i(x) + \gamma_{i-1}(x)z)(\gamma_{n-i}(x') + \gamma_{n-i-1}(x')z')$$

This follows easily on collecting the coefficients of 1, z, and z' and using condition (4) for γ. $\square$